

CS120 Project Report

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1. Project1——Physical layer

In project 1, we implemented the physical layer of Ethernet from zero.

1.1 Basic part

We choose PSK as the modulation method, as it seems the most convenient. The structure of each frame has multiple parts, including 440 bits(55 bytes) for preamble(which can ensure the following information should be received), 8 bits(1 byte) for information about the package length, 40 bits(5 byte) for Hamming code(which can correct the mistake during the transmission) and 128 bits(16 bytes) for the real information we need to transmit. Besides, we use 60 sampling points per bit.

As there are 10000 bits of information to transmit and we can transmit 128 bits in a frame, there will be 78 complete frames and 1 incomplete frame altogether. Of course, we can ensure that no problem will happen on the incomplete frame.

When dealing with the transmission part, we separated callback functions and processing functions apart. Besides, we used a lock and a queue for communication between different classes to avoid processing data within the callback function to prevent blockage.

1.2 Optional part

When dealing with the error correction part, we decided to use hamming code, which has been mentioned above. In the basic part, we used the crc(Cyclic Redundancy Check) initially, but when we consider dealing with this part, we decided to replace crc with hamming code. We implemented this part by ourselves to encode and decode our frame, which lessen the amount of mistake happen at last to a large extent.

When dealing with the OFDM part, we chose to use double carrier, with 2400Hz and 4800Hz respectively. Then we make them orthogonal to each other, which guarantees no interference within each other. Besides, we increase the number of sampling points so as to distinguish the content carried by the two carriers.

When dealing with the range challenge, we have spent several nights to tune the parameters to get a stable and highly repeatable situation. While tuning, we found that the speaker can't be too loud and the noise it sometimes generated could greatly affect the result. We borrowed microphones and speakers from others as well as bought some new ones, and continuously adjust their relative height and distance. Unluckily, when we were

tuning parameters on our own, the highest accuracy never reached 90%. However, when our project was checked, someone suggested that we should use items that were heavier and more stable to support our microphone and speaker so that they would not tremble, which may cause transmission mistake. We did so and the transmission accuracy reached 100% at the first time. We came to a conclusion that the project of computer network was just a myth.

1.3 Challenges and Opinions

It is quite a hard job for us to tune parameters. It is also a big challenge for not only us but also others who take this course, I guess, to build a framework of network transmission from zero and use brand new development tools, by which I mean, juce and projucer, although this project only contains the physical layer. I've heard about some gossip about the incredible workload involved in the projects from senior students and I admit that the first project has annoyed us for a long time, because I usually spend a lot of time on it but only get a little progress in return, while although my teammate was sometimes stuck, he could then progress sustainably. This experience does exercise us by costing us a lot of time to conduct the goals, but we learned a lot from both the hard work and the outcome, the structure of physical layer of the Ethernet, we successfully build. The brand new tools greatly broaden our abilities. But there is a bit hard for someone who has no access before, still.

2. Project2——Mac(Media Access Control) layer

In project 2, we implemented the mac layer above the physical layer.

2.1 Basic part

Initially, we would like to introduce the structure of mac frame, which includes 1bit for dest(destination address, which can mark the edge that should receive the frame), 1 bit for src(source address, which can mark the edge that send the frame. These two parts can ensure that the frame is sent to the expected destination correctly), 2 bits for type(type field, which marks the type the frame belongs, 01 for ack and 10 for frame), 8bits for idx(which marks the length of the data portion) and 256 bits for mac_layer(which is the data portion of the frame and what should be really transmitted).

What has been mentioned above is that we implemented ACK mechanism in the MAC layer, using stop-and-wait mechanism. In this project, we measured the RTT at first and set it as the resend time to speed up the transmission, which will soon be explained. On the one hand, the sender will send out messages with destination address, source address, type, data length and payload, and then wait for the ack message. If ack is not received after a fixed time slot which we set in advance, the sender will consider the transmission as failed, which means the receiver does not receive the message, and the sender will resend the message again with a limited range of times which we also set in advance. Else, which ack is received, the sender will continue to send the following messages. On the other hand, the receiver will keep listening. Once a message is detected and received, it will check the message and send the ack frame back to the sender if it is correct to inform the sender.

In the latter tasks, the consumption of time is a vital problem we should consider and requires a lot of practice and optimization. It is also a main fact that this project attaches great importance to. We used various references and move semantics to avoid copy operations, which is the main part in accelerating computation. For the interaction with the physical layer, we removed the error correction codes and replaced them with a simple checksum just to reduce computation time and it is effective. Additionally, in this project, before a data is sent, we first check whether the channel is busy. If so, the physical layer notifies the mac layer with a boolean variable, and the mac layer begins conducting backoff. It is note worthy that we should have implemented exponential backoff in the tasks theoretically, while in fact we realized both exponential backoff and linear backoff. Eventually we decided to use linear backoff because of its better performance.

When calculating the channel busy detection, we used square and power. This will promote the transmission efficiency. Besides, we initially decided to implement the sliding window algorithm, like what the lectures had shown us, in the accumulate version by using a queue. But later we found that the debugging process is tormenting and we agreed on removing that. Finally, to improve propagation efficiency, we reduced the number of sample points per bit and doubled the sampling rate.

Next part is CSMA, if the package detected contains data, we will calculate the square sum of the sampling points in a buffer and set a threshold. When the value is less than the threshold, the data will be sent.

In order to promote the time consumed, we allocated all the space in advance which may be required later and reduced the amount of operations of “copy” but replaced them with “move” so that the efficiency could be guaranteed.

2.2 Optional part

The optional part of this project only cares about the time performance and the number of nodes, which is, in other words, transmission speed and number of devices.

About the speed part, we’ve mentioned a lot in the above parts, including the challenges we faced and the solutions we came up. So I won’t repeat that again.

Besides, there is not enough time for us to get more nodes and then implement the optional part.

As a result, we did do a lot in promoting the transmission speed and it’s a hard job.

2.3 Challenges and Opinions

In the first 2 projects, we are now a bit familiar with the Juce. It is crucial to read the Juce low-level source code from zero and correctly implement our ideas and the tasks as expected. It's important to understand its low-level to avoid some bugs and errors, which can sometimes cost a lot of time to debug or find for help.

Besides, in this project, the packet length should not be too long, and alignment issues should be taken into consideration.

It is still quite a hard job for us to tune parameters in this project. But we are a bit skilled after the hell-like experience in the last project. This time, how to speed up is the main problem. Our efforts can be seen above, in the “basic part”. to accelerating the speed we conduct a lot of practice and optimization, remove and add a lot of function. This time

we failed to realize the inference part, which is said to be extremely hard. We've told that the last 2 projects will be a little easier and we hope that it is true.

3. Project3——IP layer

In project 3, we implemented the IP layer on the basis of the projects before.

3.1 Basic part

To begin with, I have to mention that this project is based on the former 2 projects of physical layer and mac layer. Then we began to implement the IP layer. We used a virtual network interface card and transmitted on it.

We use the command "New-NetNat" in Windows to realize NAT in node 2 which we set the internal ip as a subnet, then windows will perform the NAT operation.

Routers analyze the destination addresses of incoming packets and forward them to the next hop according to routing rules. By checking the IP headers, they verify whether the destination meets certain criteria. If it does, the packet is forwarded to the appropriate next hop.

3.2 Optional part

We implemented a network interface card and successfully realized the task 9,10,11.

As we failed to find a well-performed C++ package manager and it's difficult to use a library in VS, we decided to use zmq to build the communication between python and C++.

3.3 Challenges and Opinions

In this project, multiple terminals are required. As we all know, tuning parameters is quite a hard job and it is time-consuming. When we almost completed the tasks required, we had to collect all the devices required for a certain period of time and then managed to tune the parameters and do the last debug jobs. But as SISTors, it is hard for us to donate our computers just to do a group job, while we are stuck to this mission and unable to freely juggle other tasks.

Besides, when we are checked in the ordered classroom, we noticed that the wifi there isn't well enough because when we were trying our best to show what we have done, the wifi broke and we can do nothing but to wait for recovery. And during the checking period of the following tasks, the wifi of the other terminal broke again, which confused us a lot. I want to propose if the checking phase can be conducted in a place with a more stable wifi condition.

4. Project4——Above IP

4.1 Basic part

On the basis of Project3, we add the selection mechanism of DNS and TCP packages.

The DNS protocol is implemented to resolve domain names to IP addresses. We use scrapy library to filter the DNS package, we check whether the port is equal to 53 and check whether the domain is equal to "example.com" or "baidu.com".

We use the scapy library to filter the tcp package, we record the original seq and change it to the random number we generate at first, then when we receive the replay, we calculate the corresponding index of ack, we change the ack index to the correct number

4.2 Challenges and Opinions

In this part, we'd like to do an overall summary of this course and the 4 projects.

Having undergone the course of CS120, we've learned a lot from both the class content and the experience of managing to conduct all the projects. Prof. Yang is experienced and has a sense of humor. Knowledge can be transmitted in our brain in a relatively easy way as long as we are diligent enough to take the morning-8-class. Meanwhile, Prof. Yang is welcome to all our questions and able to answer that.

However, when it comes to the projects, it's hard to feel convenient. As what is mentioned in Project 1, the projects are extremely time-consuming. Besides, at the beginning of the semester, it's hard to get to learn the development tools this course required, namely ASIO and projuder. It is also hard in the very beginning to make up the framework of the tremendous tasks. We propose that if there could be some courses which can teach us how to use the tools so that we won't be so desperate when facing the first project.